```
G = ([v1, v2, v3, v4], [(v1,v2),(v1,v3),(v2,v4)])
```

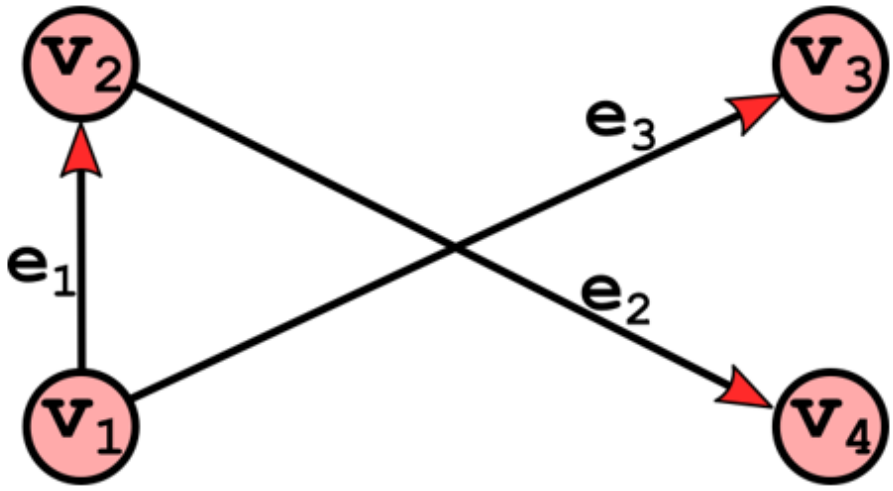


Figure 14.2 An example of a (directed) graph.

An edge whose source and destination are the same is called a *loop* (see figure 14.3); *simple graphs* can't have any loop, nor they can have multiple edges between the same pair of nodes; conversely, *multigraphs* can have any number of edges between two distinct vertices. Both simple graphs and multigraphs can be extended to permit loops.